



Technical Service Bulletin: TSB170122

Released Date: 28-Jan-2026

White Residue / Diesel Exhaust Fluid (DEF) Staining On Exhaust

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- ISX15 CM2350 X101
- X15 CM2350 X114B
- X15 CM2350 X116B
- X15 CM2350 X123B
- X15 CM2450 X124B
- X15 CM2450 X134B
- X15 CM2450 X142B
- X15 CM2450 X155B

Issue

Symptom:

- White residue on exhaust stacks and/or exhaust
- White staining on exhaust stacks and/or exhaust
- White powder and/or residue on equipment frames



Figure 1, White Staining on Exhaust Stack.



Figure 2, White Powder on Frame.

Root Cause:

- Urea **not** being decomposed/atomized to gaseous state completely for any of the following reasons

- Presence of diesel exhaust fluid (DEF) deposits in aftertreatment decomposition tube or mixer
- Overdosing aftertreatment DEF dosing valve
- Incorrect concentration of DEF
- However, there may be nothing wrong with DEF system and residue, staining, or powder might be more noticeable for applications that operate with lower aftertreatment temperature less than 250°C [482°F].
- Presence of white staining / DEF staining on exhaust stack does **not**
 - Impact performance of engine and aftertreatment.
 - Impact durability of aftertreatment or exhaust system.
 - Increase DEF consumption.

Resolution

- If any fault codes related to aftertreatment and/or DEF system are active, troubleshoot fault codes using corresponding troubleshooting tree.
- If there is a complaint of excessive DEF consumption, follow the appropriate Excessive Diesel Exhaust Fluid (DEF) Consumption troubleshooting.

Table 1, Publications Affected					
Manual Type	Engine	Bulletin Number	Procedure Title	Procedure	Section
Service Manual	X15 CM2350 X114B	5411181	Exhaust System Diagnostics	Refer to Procedure 011-056 (/qs3/pubsyss2/xml/en/procedures/493/493-011-056.html)	11
	X15 CM2350 X116B	5411186		Refer to Procedure 011-056 (/qs3/pubsyss2/xml/en/procedures/493/493-011-056.html)	
	ISX15 CM2350 X101	4310641		Refer to Procedure 011-056 (/qs3/pubsyss2/xml/en/procedures/173/173-011-056-tr-prix13.html)	

Document History

Date	Details
2017-11-27	Module Created
2025-9-18	Updated Product Affected.
2025-10-20	Reviewing tsb to reset status on it to active. No changes made.
2026-1-28	Added X124B, X134B, and X142B to Product Affected list.

